

mstsa: A Python Library for Information-Theoretic and Statistical Analysis of EEG Microstate Sequences

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DOI: [N/A](#)

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Submitted: 01 January 1970

Published: 01 January 1970

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Summary

EEG microstate analysis decomposes a multichannel EEG recording into a discrete symbolic time series: each sample is assigned to one of a small number of prototypical scalp topographies (the microstates), typically at sampling rates of 250–1000 symbols per second ([Lehmann et al., 1998](#); [Michel & Koenig, 2018](#); [Pascual-Marqui et al., 1995](#)). Once the segmentation step is complete, the scientific question shifts to the temporal structure of the resulting sequence: Is it Markovian? Stationary? How complex is it, and how does that complexity compare to a random null model?

mstsa (Microstate Time Series Analysis) is a Python library that addresses these questions. It provides a comprehensive set of information-theoretic, Markov-chain, and statistical tools for the analysis of discrete symbolic sequences, with EEG microstate data as the primary application. The library exposes six modules:

- **mstsa** (core): Shannon, Rényi, and Tsallis entropies; block, topological ([Adler et al., 1965](#)), and dispersion ([Rostaghi & Azami, 2016](#)) entropies; entropy rate and excess entropy ([Cover & Thomas, 2006](#)); auto-information function (AIF) and partial auto-information function (PAIF) ([von Wegner et al., 2016](#); [von Wegner, 2018](#)); active information storage (AIS) ([Lizier, 2014](#)); sample entropy for discrete sequences, using exact-match comparison and a runs-tracking algorithm that computes estimates for all template lengths in a single $O(N^2)$ pass ([Lake et al., 2002](#); [Richman & Moorman, 2000](#)); Lempel–Ziv complexity ([Kaspar & Schuster, 1987](#); [Lempel & Ziv, 1976](#)); detrended fluctuation analysis (DFA), rescaled-range (R/S) analysis, and diffusion entropy analysis (DEA) ([Scafetta & Grigolini, 2002](#)), applicable to both binary-partition random walks and to indicator functions (one-hot encodings) of individual microstate classes ([Hurst, 1951](#); [Peng et al., 1994](#)); microstate duration, occurrence, and coverage; power spectral densities of indicator functions; and random-walk construction from binary partitions ([Van De Ville et al., 2010](#); [von Wegner et al., 2016](#)).
- **markov**: Conditional and joint transition probability matrices, stationary distribution, continuous-time generator matrix, AIF and sample entropy under first-order Markov models ([von Wegner, 2026](#)), surrogate sequence generation ([von Wegner et al., 2025](#)), nearest-reversible Markov chain projection ([Nielsen & Weber, 2015](#)), and detailed-balance statistics ([Lynn et al., 2021](#); [Roldán & Parrondo, 2010](#)).
- **stats**: Likelihood-ratio (G) tests for zero-, first-, and second-order Markovianity; j- and jk-homogeneity (stationarity); conditional homogeneity; transition-matrix symmetry ([Kullback et al., 1962](#)); geometric sojourn-time distributions; sojourn-time distribution fitting that contrasts power-law, exponential, stretched-exponential, and lognormal candidates via the powerlaw package ([Alstott et al., 2014](#); [Clauset et al., 2009](#)); and a permutation-based multi-subject transition syntax test ([Lehmann et al., 2005](#)).
- **acd**: Autoregressive Conditional Duration (ACD) models ([Engle & Russell, 1998](#)) of sojourn-time sequences, with exponential, Weibull, or lognormal innovations, AIC/BIC-based order selection, a Ljung–Box residual autocorrelation test, ACD surrogate-sequence

simulation, and ACF/PACF diagnostic plots — quantifying serial clustering between consecutive sojourn durations, complementary to the marginal-distribution fitting in stats.

- **eeg:** Wackermann/Palus multichannel EEG complexity measures — effective dimensionality Ω , effective voltage Σ , generalised frequency Φ , and Palus complexity coefficient (Paluš et al., 1992; Wackermann, 1996); and continuous sample entropy (sample_entropy_cont), using tolerance-based matching (Lake et al., 2002; Richman & Moorman, 2000).
- **microsynt:** Python implementation of the Microsynt syntactic analysis method (Artoni et al., 2023).

Computationally demanding routines are accelerated by Numba JIT compilation with persistent disk caching (Lam et al., 2015) and optional CFFI-linked C extensions for block entropy, Lempel–Ziv complexity, and continuous and discrete sample entropy.

Statement of Need

EEG microstate research spans five decades and has produced a rich body of results linking microstate dynamics to cognition, sleep, anaesthesia, and neurological disorders (Michel & Koenig, 2018). The standard analysis pipeline has two stages: (1) identifying the prototypical maps by clustering, and (2) analysing the resulting symbolic sequence.

Stage 1 is well covered by existing software. CARTOOL (Brunet et al., 2011), the EEGLAB microstate toolbox (Poulsen et al., 2018), MICROSTATELAB (Nagabhushan Kalburgi et al., 2023), and the Python library Pycrostates (Férat et al., 2022) all provide mature clustering and segmentation tools. Pycrostates in particular integrates cleanly with the MNE-Python ecosystem and supports both subject- and group-level analyses.

Stage 2 is far less systematically served. The conventional summary statistics — mean duration, occurrence rate, fractional coverage, and transition probabilities — have been available for decades, but they characterise the *marginal* and *first-order* structure of the sequence only. The information-theoretic characterisation that reveals *higher-order* temporal dependencies — entropy rate, excess entropy, auto-information structure, and Lempel–Ziv complexity — and, critically, their Markov-chain reference values that establish significance baselines, has only been formalised recently (von Wegner et al., 2017; von Wegner, 2026; von Wegner & Laufs, 2024). These methods are currently scattered across individual publications. An earlier Python package implementing an initial subset of these methods was published alongside the foundational information-theoretic analyses (von Wegner & Laufs, 2018b); mtsa extends and maintains that work with a substantially expanded function set, C-accelerated implementations, and Numba JIT compilation.

mtsa fills this gap. It is the only Python library that (i) implements the full set of information-theoretic complexity measures for symbolic sequences alongside their first-order Markov-chain predictions, enabling direct quantification of non-Markovian excess structure; (ii) provides the Kullback-information likelihood-ratio tests for Markovianity, stationarity, and transition-matrix symmetry (Kullback et al., 1962); (iii) includes a permutation-based multi-subject test for non-random microstate transition syntax (Lehmann et al., 2005); and (iv) models serial dependence between consecutive sojourn durations directly, via Autoregressive Conditional Duration models (Engle & Russell, 1998), rather than only characterising each duration's marginal distribution in isolation. The ability to compare empirical measures to analytically derived Markov-chain benchmarks — rather than relying solely on computationally intensive surrogate ensembles — is a distinctive feature of the library (von Wegner, 2026).

The library is standalone: it does not require MNE-Python and operates on any integer-valued symbolic sequence. This makes it applicable beyond EEG to any domain in which discrete time series arise, such as fMRI parcellation sequences, animal-behaviour state data, or genomic symbol strings. An earlier version (0.2.0) has been used in a series of peer-reviewed publications

characterising microstate dynamics during resting state, propofol anaesthesia, and epilepsy (Hermann & others, 2024; von Wegner et al., 2016, 2017; von Wegner, 2026; von Wegner & Laufs, 2018a, 2024).

Acknowledgements

This work did not receive external funding.

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